

Classification of Drone Type Using Deep Convolutional Neural Networks Based on Micro-Doppler Simulation

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Abstract – Monitoring drones has become an increasingly significant area of study for surveillance and safety purposes. The use of radar is one of the most feasible approaches to detecting a drone as RF waves propagate with a low attenuation constant. In this study, we suggest classifying drones based on the micro-Doppler signatures in the spectrogram when observed by radar. We simulate micro-Doppler signatures from various drones and investigate the feasibility of classifying the type of drone. A deep convolutional neural network is suggested as a classifier and its classification accuracy is reported.

Index Terms — Micro-Doppler signatures, Deep Convolutional Neural Networks, Drone tracking.

1. Introduction

The classification and identification of drones beyond the visual range has become an increasingly valuable area of study for safety and security purposes. A single drone or group of drones are a potential threat to crowds or important facilities related to power, defense, and security. In addition, drones can be used to invade privacy by gathering unauthorized information. To reduce such risks, detection of drones at a distance is required. To detect a drone at a distance, it is desirable to use a radar system because electromagnetic waves have relatively low attenuation with distance and the radar system has a broader beam with which to search a larger area compared to an optical system [1]. To discern drones from other airborne objects such as birds or debris, the characteristics of drones should be investigated.

One of the most significant characteristics of flying drones is the existence of micro-Doppler signatures [2]. Unlike a solid body, the moving parts of an object produce modulated Doppler components called micro-Doppler. The modulated Doppler signature is presented as added components to the Doppler signature of the drone body. Due to the rotating blades, frequency modulated components, which are quite revealing, are produced. To observe the time-varying micro-Dopplers, the received FMCW should be processed by joint time-frequency analysis. Because the micro-Doppler signatures depend on the number of rotors, rotor speed, and orientation of the

drone, the analysis of micro-Doppler signatures can provide information pertaining to the drone [3].

In this study, we propose to classify drones based on the deep-learning algorithms after transforming the micro-Doppler signatures to images through the joint time-frequency analysis. We will use the short-time Fourier transform and the Wigner-Ville distribution for the joint time-frequency analysis. For this study, we simulate micro-Doppler signatures from diverse drones by changing the number of rotors, rotor speed, and orientation. For the classifier, we propose to use a deep-learning algorithm that has been proven to be very powerful in image recognition. Specifically, we employ the Deep Convolutional Neural Network (DCNN) which is composed of the convolution layer, pooling layer, and fully connected layer. The DCNN is trained with the simulated data through the gradient descent method. The optimal hyperparameters are found by a heuristic search and the classification accuracy is reported.

2. Simulation of Micro-Doppler Signatures from the Drone

To simulate the Doppler signature from the drone, the different parts of the drone body are modeled as primitive shapes. For example, cylinders, ellipsoids, and spheres whose radar cross-sections (RCS) are well characterized analytically are used to describe the drone structure [4]. To simulate the micro-Doppler signatures of the drone, the rotating blade is modeled as a group of point scatterers. As the maximum Doppler frequency is generated from the edge of the blade, the blade is divided into several plates and each plate is modeled as a point scatterer with a corresponding radar cross-section (RCS). Because the Doppler frequency originates from the radial velocity, it is calculated based on radar position, drone orientation and operating frequency. In the simulation, an electric field incident on the moving drone is assumed by

$$E_i(r) = \frac{A}{r} e^{-j\frac{2\pi}{c}r} \quad (1)$$

A is a constant related to the transmission power, and r is the distance from the transmitter to the drone. If we assume a single scattering model, the total scattered field can be generated by the complex sum of all K body parts as in the following equation.

$$Es(t) = \frac{A}{\sqrt{4\pi}} \sum_{b=1}^K \sqrt{\sigma_b(t)} \frac{e^{-j\frac{2\pi}{c}2r_b(t)}}{r_b(t)^2} \quad (2)$$

The total time domain of the echo can be found as a superposition of Doppler signals from each scatterer. We can simulate the spectrogram of the micro-Doppler signatures of the drone in diverse scenarios. The rotor speed, the number of rotors, and the orientation of the rotors have been varied to emulate practical situations. The micro-Doppler signatures can be used for classification purposes [5].

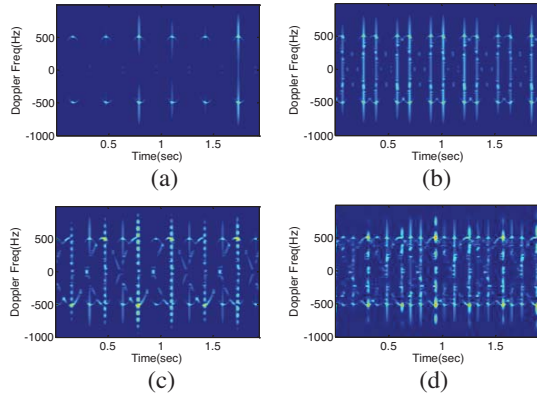


Fig. 1. Simulated micro-Doppler signatures from several drones, (a) single rotor, (b) two rotors, (c) three rotors, and (d) four rotors.

3. Classification Using Deep Convolutional Neural Networks

Deep learning, a subset of machine learning, has recently gained much attention and has made tremendous advances in several areas of application such as speech recognition, image recognition, and natural language processing. The difference between conventional supervised learning and deep learning is that deep learning uses multiple layers to extract and generalize features and construct the classification boundary at the same time. Even though deep learning is not a new concept, it has only been made practical recently owing to the achievement of excellent empirical performance in several applications. High-performance computing processors such as graphics processing units and the continuation of algorithmic innovation were the main drive for deep learning.

Because DCNNs have shown great success in image classification, it is natural that we employ one to discern the drone micro-Doppler signatures. The practical DCNN consists of several layers. The first layer is the convolution filter. The convolution filter is the main part of DCNN. With a sliding window, it is applied to small local fields of input data to extract a feature. Because each layer employs multiple convolution filters in parallel, diverse features can be constructed.

After nonlinear activation functions are imposed on the convolution filter output, the next component is the pooling

layer that diminishes the data size for the robust translation of input data. The last stage consists of the dropout layer, fully-connected layer and classifying layer.

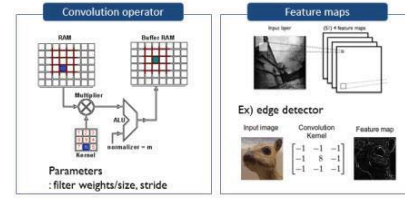


Fig. 2. Operation of convolutional filter.

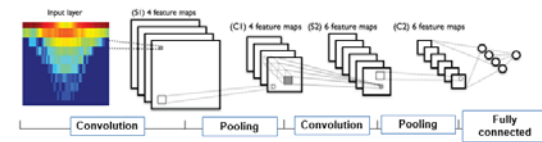


Fig. 3. Operation of convolutional neural networks.

4. Classification Results

For the training of the DCNN, we have simulated 500 spectrograms by varying the number of rotors, the rotor speed, and the orientation of the drone. With the simulated spectrograms, the DCNN has been trained with a gradient descent method. The hyperparameters of the DCNN are heuristically optimized. The final classification accuracy with a set of 20% of test data was 93%.

5. Conclusion

In this study, we investigated the feasibility of classifying types of drones with simulated micro-Dopplers with a DCNN. The developed simulator could produce micro-Doppler signatures which could be used to train a DCNN. Based on the high classification accuracy, this study can serve as a pilot study before the actual field measurements.

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